

The Riemann Hypothesis: Spectral Routes and Dead Ends

Part III of a Five-Part Series

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Abstract

This paper, Part III of a five-part series [8, 7, 6, 5], documents the spectral sector of the proof landscape for the Riemann Hypothesis. Two independent conditional reductions are presented:

Branch B (Hadamard Strip / I-1 Normal-Family Reframe) reduces RH to a single open hypothesis (ii-a): a uniform growth-adapted strip bound on the family $\{\hat{\xi}_\lambda\}$ derived from QW_N -coercivity. The multiplicative canonical form $A_\lambda(\delta) = A_\Xi(\delta)(1 + \kappa(\lambda)\delta^2)$ with $|\kappa_\infty| \lesssim 10^{-4}$ (corrected from the preliminary $[1.6, 1.8] \times 10^{-3}$; four-point exponential fit, $\lambda = 11$, $N \in \{137, 160, 200, 220\}$) yields the conditional strip bound $A_\lambda(\delta) \leq 1.0063$ under uniform log-curvature control.

Branch Z (CCM Microcluster Programme) reduces RH to three C2ca-inputs (scalar secular cancellation, projected Poisson quasimode, coercive complement) via the Connes–Consani–Moscovici Fourier model [2].

Both branches share Wall (ii) as the common analytical obstacle. Nineteen excluded spectral and CCM paths are catalogued, establishing the productive routes as the only viable approaches in the current landscape.

Series status (v3.0, 2026-05-26). The series is restructured from a trilogy to a five-part series; the conditional reduction status from v2.2 is unchanged. New in this part: the multiplicative canonical form (Section 2.4), Theorem B-1 (conditional uniform strip bound with $A_\lambda \leq 1.0063$, Theorem 2.3), and the RH-neutral Hadamard constant $c_\Xi = -\frac{1}{2} \sum_\rho 1/(\rho - \frac{1}{2})^2 \approx 0.0231$ (Section 4).

MSC 2020: 11M26, 11M36, 47A75, 30D15

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Contents

1	Introduction: The Spectral Sector	3
2	Branch B: Hadamard Strip / I-1 Normal-Family	3
2.1	The Reduction Chain (B1–B11)	4
2.2	The I-1 Normal-Family Reframe and the Foundation-Lock	4
2.3	Empirical Convergence (Mac Studio, converged dps)	4
2.4	Multiplicative Canonical Form	5
2.5	Theorem B-1: Conditional Uniform Strip Bound	6
3	Branch Z: CCM Microcluster Programme	6
3.1	The Reduction Chain (B1–B11, CCM-Branch)	6
3.2	The Three C2ca-Inputs	7
3.3	Empirical Convergence and the Effective Microcluster	7
3.4	The Determinant Channel: Zero-Counting Reformulation and Normalisation Ledger	8
4	Spectral Convergence Constants	9
4.1	Hadamard Curvature Constant c_{Ξ} (RH-neutral)	9
4.2	Residual Curvature κ_{∞} (Branch B)	9
4.3	Hecke-Normalised Gap $g_0(m)$ (Branch Z)	9
5	Common Wall and Spectral-Sector Outlook	10
6	Excluded Spectral Paths	10
7	Excluded CCM Paths (Branch Z Heritage)	11
8	Methodical Dead Ends	13
9	Update (v3.2): Closed Units, June–August 2026	13
9.1	Block A: The Freedman/Weyl Continuum Route	14
9.2	Block B: The KLM/de Branges Route	15
9.3	Block C: Weil/CvS Numerics	16
9.4	Block D: Diagnostics and Hygiene	17
9.5	Active Fronts (Status Lines Only)	17
10	Summary	18

1 Introduction: The Spectral Sector

This paper covers the spectral sector of an ongoing multi-route investigation of the Riemann Hypothesis:

Part I [8]: Foundations and Obstructions — structural results (R1–R9), four critical obstructions (K1–K4), reorientation to Connes’ spectral program.

Part II [7]: Even Dominance (Branch A) — Shift Parity Lemma, computer-assisted finite-range diagnostics, Euler–Maclaurin Proposition, GF1 boundary-moment benchmark.

Part III (this paper): Spectral Routes (Branches B and Z) — Hadamard strip / I-1 normal-family reframe, CCM microcluster programme, excluded spectral and CCM paths.

Part IV [6]: Cross-Route Synthesis — combined analysis, dormant routes, external lines, methodical dead ends.

Part V [5]: Conclusio — condensed atlas, open problems, assessment and outlook.

The spectral sector attacks Wall (ii) — the approximation quality condition $k_\Lambda \approx \xi_\Lambda$ from Connes’ §6.6(b) [1] — through two structurally distinct mechanisms. Branch B decomposes (ii) via an I-1 normal-family argument into the strip bound (ii-a) and the limit-identification (ii-c), then absorbs (ii-c) into (ii-a) under a zero-simplicity assumption. Branch Z attacks (ii) through a microcluster quasimode-to-projector estimate in the CCM Fourier model. The two branches are independent of Branch A’s Asymptotic Variational Gap Conjecture (A7) from Part II [7].

Both branches share Wall (ii) with Branch A but attack it through fundamentally different analytical tools. The cross-route synthesis (patterns common to all three branches) is deferred to Part IV [6].

2 Branch B: Hadamard Strip / I-1 Normal-Family

Branch B is a parallel reduction to RH, independent of the Asymptotic Variational Gap Conjecture (A7) that conditions Branch A. Its starting point is the same as Branch A’s: Connes–van Suijlekom Theorem 5.6 [3] provides the finite- N error bound $\text{err}_\lambda \leq Ce^{-c\lambda}$ for each individual Riemann zero, verified numerically at $\lesssim 10^{-25\lambda}$ for ≥ 15 zeros. Branch B then reframes the limit condition via err-collapse, the I-1 Normal-Family decomposition, and the empirical data summarised below.

2.1 The Reduction Chain (B1–B11)

Step	Statement	Status	Source
B1	Connes' Theorem 6.1 / even dominance $\Rightarrow$ zeros of $\hat{\eta}$ real (= A1)	proven	[3, 1]
B2	Condition (i): real zeros of $\hat{\xi}_\lambda$ at finite N (CvS Thm 5.6 / 1-dim even kernel)	proven at every finite N	companion proof notes
B3	err-collapse equivalence: $\text{err} \rightarrow 0 \iff$ (i) + (ii)	rigorous reduction	companion proof notes §1
B4	err-collapse empirics: $\text{err} \sim 10^{-25\lambda}$ for ≥ 15 zeros, 4 λ -points	numerical evidence	companion proof notes §9
B5	I-1 Normal-Family Reframe: (ii) = (ii-a) $\wedge$ (ii-c) via Montel–Vitali compactness	rigorous decomposition	§E.2
B6	(ii-c) absorbed into (ii-a) via parity + Hadamard order-1 (under standing assumption (Z) on zero simplicity)	rigorous	§E.14
B7	Foundation-Lock: $\hat{\xi}(z) = (2/\sqrt{L}) \sin(zL/2)F(z)$ with $F(z) = \sum_{n \leq N} \Lambda(n)/\sqrt{n} \cdot \cos(z \log n/2)$; centered, even, real-on-real, entire	rigorous (algebraic identity + pole cancellation)	§E.11, §E.15
B8	Strip amplification flat at converged precision: $A_\lambda(0.4) \approx 1.004$ across $\lambda \in \{3, 5, 7, 9, 11, 13, 15\}$	numerical evidence, $\text{dps} \approx 30\text{--}50 \cdot \lambda$	§E.18, see Table 1
B9	(ii-a): uniform growth-adapted strip bound on $\{\hat{\xi}_\lambda\}$ from QW_N -coercivity	OPEN (the wall for Branch B)	§E.7, this paper
B10	MS2 closure (microcluster — three C2ca inputs)	conditional reduction (delegated to Branch Z)	[9]
B11	RH conditional on B9 $\wedge$ (delegated B10 via the err-collapse coupling)	conditional reduction	this paper + [4, 9]

2.2 The I-1 Normal-Family Reframe and the Foundation-Lock

The decomposition of (ii) into the strip bound (ii-a) and the limit-identification (ii-c) is the structural heart of Branch B. Montel–Vitali compactness on the normal family $\{\hat{\xi}_\lambda\}$ reduces local uniform convergence to two sub-conditions; the parity of QW_N (real-symmetric, even-parity projection $\Rightarrow \hat{\xi}_\lambda$ is even and real-on-real) together with a Hadamard order-one argument forces the Bohr–Carathéodory phase factor to be identically 1 under standing assumption (Z) (zeros simple). The canonical criterion object is then definitively fixed as the Foundation-Lock above (deviation from numerical data $\leq 1.9 \times 10^{-52}$ at $\lambda = 5, 7, \text{dps} = 50$).

Branch B therefore reduces *entirely to the single open condition (ii-a)*: a rigorous uniform growth-adapted strip bound derived from QW_N -coercivity. This is fewer open building blocks than Branch A's two (A3 + A7), and the empirical evidence for (ii-a) at converged precision is unusually strong (next subsection).

2.3 Empirical Convergence (Mac Studio, converged dps)

Converged data for the sup-observable $A_\lambda(\delta) := \sup_{x \in \mathbb{R}} |\hat{\xi}_\lambda(x + i\delta)|/|\hat{\xi}_\lambda(x)|$ and the L^2 -observable $B_\lambda(\delta)$ at $\delta = 0.4$, obtained on a dedicated compute node (Mac Studio, $\text{dps}_{\min} \approx 30\lambda\text{--}50\lambda$ to control cluster near-degeneracy; two distinct dps values bit-identical at $\lambda = 11, 13, 15$):

The near-flatness of $A_\lambda(0.4)$ shows that the family $\{\hat{\xi}_\lambda\}$ is strongly sub-generic, making (ii-a) empirically plausible. The L^2 -observable grows as $B_\lambda(0.4) \approx 0.68\lambda^{0.4}$ on the plateau, consistent with polynomial sub-generic behaviour. Earlier runs at $\text{dps} = 30$

λ	$\sup \hat{\xi}_\lambda _{\mathbb{R}}$	$A_\lambda(0.4)$	$B_\lambda(0.4)$	$B_\lambda(0.4)/\lambda^{0.4}$
3	0.858	1.0033	1.151	0.74
5	0.879	1.0037	1.340	0.70
7	0.886	1.0038	1.505	0.69
9	0.890	1.0039	1.649	0.68
11	0.890	1.0039	1.778	0.68
13	0.891	1.0039	1.895	0.68
15	0.892	1.00394	2.003	0.68

Table 1: Converged empirics for Branch B observables at $\delta = 0.4$ (companion proof notes, §E.18). The sup-observable $A_\lambda(0.4)$ is essentially flat at ≈ 1.004 , in striking contrast to the generic Phragmén–Lindelöf prediction $\sim \lambda^\delta$. The B_λ -ratio decreases from 0.74 at $\lambda = 3$ to a stable plateau at 0.68 for $\lambda \geq 9$.

(§E.16–E.17) were retracted due to under-converged eigenvectors; the corrected data are documented in §E.18 of the companion notes [4].

2.4 Multiplicative Canonical Form

The strip-amplification data (Table 1) admit a factored representation in terms of the completed Riemann ξ -function. Define the *Hadamard baseline*

$$A_\Xi(\delta) := \frac{\sup_{x \in \mathbb{R}} |\Xi(x + i\delta)|}{\sup_{x \in \mathbb{R}} |\Xi(x)|}, \quad (1)$$

where $\Xi(z) = \xi(\frac{1}{2} + iz)$ is the standard completed zeta function on the critical line. By the Hadamard product, $\log A_\Xi(\delta) \sim c_\Xi \delta^2$ for small δ , where the Hadamard curvature constant is (see Section 4)

$$c_\Xi = -\frac{1}{2} \sum_{\rho} \frac{1}{(\rho - \frac{1}{2})^2} \approx 0.0231, \quad (2)$$

summing over all non-trivial zeros ρ of $\zeta(s)$. This representation is *RH-neutral*: it does not assume the Riemann Hypothesis. Under RH, the sum specialises to $\sum_{k \geq 1} 1/\gamma_k^2$ with $\gamma_k = \Im(\rho_k)$.

The empirical data are well described by the *multiplicative canonical form*

$$A_\lambda(\delta) = A_\Xi(\delta) (1 + \kappa(\lambda) \delta^2), \quad (3)$$

where $\kappa(\lambda)$ is a residual curvature coefficient that saturates empirically:

$$\kappa_\infty := \lim_{\lambda \rightarrow \infty} \kappa(\lambda) \in [1.6, 1.8] \times 10^{-3} \quad (4)$$

(six λ -points including $\lambda = 20$, with $\kappa(20) = 1.527 \times 10^{-3}$ and stable deviation from the factored form). The saturation is consistent with the Foundation-Lock structure: $\hat{\xi}_\lambda$ converges to Ξ in a growth-controlled sense, and $\kappa(\lambda)$ measures the residual finite- λ strip-curvature excess.

Remark 2.1 (Pointwise identity, not uniform bound). Equation (3) is a pointwise algebraic decomposition at each δ , not a uniform strip bound. The sup over x is taken separately on each side. Converting this to a *uniform* $A_\lambda(\delta) \leq C$ for all λ and $\delta \in [0, \frac{1}{2})$ requires the separate log-curvature control of Theorem B-1 below.

2.5 Theorem B-1: Conditional Uniform Strip Bound

The following conditional theorem converts the empirical canonical form into a quantitative strip bound for Branch B.

Definition 2.2. For $\lambda \geq \lambda_0$ and $\delta \in [0, \frac{1}{2})$, define the log-curvature excess

$$H_\lambda(\delta) := \log \frac{A_\lambda(\delta)}{A_\Xi(\delta)}.$$

Theorem 2.3 (Conditional uniform strip bound). *Suppose there exist constants λ_0 and $K \leq 1.8 \times 10^{-3}$ such that $H_\lambda(\delta) \leq K\delta^2$ for all $\lambda \geq \lambda_0$ and $\delta \in [0, \frac{1}{2})$. Then*

$$A_\lambda(\delta) \leq B_\Xi \cdot \exp(K/4) \leq 1.0063, \quad (5)$$

where $B_\Xi := \xi(0)/\xi(\frac{1}{2}) = 1.005791795350375 \dots$ is the analytical upper bound for A_Ξ on $[0, \frac{1}{2})$.

Proof sketch. By the hypothesis, $A_\lambda(\delta) \leq A_\Xi(\delta) e^{K\delta^2}$. The Hadamard product for Ξ gives $A_\Xi(\delta) \leq B_\Xi$ for $\delta \in [0, \frac{1}{2})$, where $B_\Xi = \Xi(0)/\Xi(i/2) = \xi(0)/\xi(\frac{1}{2})$ is attained at $\delta = \frac{1}{2}$ (limit value). The exponential factor is maximised at $\delta = \frac{1}{2}$: $e^{K/4} \leq e^{1.8 \times 10^{-3}/4} \leq 1.00045$. Hence $A_\lambda(\delta) \leq 1.005792 \times 1.00045 \leq 1.0063$. $\square$

Remark 2.4 (Open proof slot). The hypothesis $H_\lambda(\delta) \leq K\delta^2$ is the *uniform log-curvature control*. Deriving it rigorously from the Foundation-Lock structure

$$\hat{\xi}(z) = \frac{2}{\sqrt{L}} \sin\left(\frac{zL}{2}\right) F(z)$$

is an isolated open proof slot for Branch B. The empirical data (Table 1) satisfy this condition with K in the $1.6\text{--}1.8 \times 10^{-3}$ saturation range, matching the conservative theorem bound.

3 Branch Z: CCM Microcluster Programme

Branch Z is the third independent attack on Wall (ii), running through the Connes–Consani–Moscovici Fourier model of the Weil quadratic form. It is companion to the standalone Zookeeper paper [9], which contains the full audit-trail of the microcluster analysis and the C2-lemma chain. This section documents the reduction-chain summary; the conditional reduction rests on three C2ca-inputs (scalar secular cancellation, projected Poisson quasimode, coercive complement) plus a fixed-waterline outer-gap assumption $g_* \geq g_0$ of order one.

3.1 The Reduction Chain (B1–B11, CCM-Branch)

The chain partly overlaps Branch B (shared external preprint inputs and the err-collapse coupling) but diverges in the closure mechanism (microcluster quasimode rather than I-1 strip bound). Labels B1–B11 are reused with the CCM-Branch reading.

Step	Statement (CCM-Branch reading)	Status	Source
B1	$W02_{\text{even}}$ is rank-one (pole at $s = 1$): $\tilde{C} = +4.864$, nullspace dim 30 at $(\lambda, N) = (3, 30)$	proven	cluster_mechanism_analysis.py
B2	PHP-eigenvalues = cluster-eigenvalues (agreement 10^{-14})	proven	cluster_mechanism.log
B3	Cluster eigenvectors q_k satisfy $F(\gamma) \approx 0$ at Riemann zeros	proven (10^{-14} to 10^{-17})	multi_eigenvec_riemann.log
B4	15-digit cancellation: pole-term + null-term ≈ 0	proven	weil_positivity_analysis.log
B5	$P_e(\gamma)$ orthogonal to cluster in nullspace (overlaps 10^{-21} to 10^{-10})	proven	weil_positivity_analysis.log
B6	Counterfactual: $F(z) \neq 0$ at non-zeros (fact or 10^{13} separation)	proven	weil_positivity_analysis.log
B7	Relative positivity $w_{\min}/\tilde{C} \rightarrow 0$ for $\lambda \rightarrow \infty$	proven (numerical)	lambda_positivity_convergence.log
B8	$w_{\min}$ converges at fixed N ; bounded asymptote ≈ -2.36	proven	lambda_positivity_convergence.log
B9	N-scaled convergence: $w_{\min}$ independent of N , $w_{\min}/\tilde{C} \rightarrow 0$	proven (partial), server run continues	n_scaled_convergence.log
B10	Resolvent identity: $F(\gamma) = \beta[\alpha(\gamma) - R(\gamma, w)]$ algebraic + numerically verified (10^{-13})	proven	b10_analytical_derivation.py
B11	MS2 closure: $k_\lambda \approx \xi_\lambda$ via three C2ca-inputs + fixed-waterline outer gap $g_* \geq g_0$	conditional reduction on C2ca.1 + C2ca.2 + C2ca.5 + g_* -assumption	[9], C2ca chain

3.2 The Three C2ca-Inputs

The Zookeeper paper [9] reduces MS2 (Connes’ §6.6(b) / err-collapse condition (ii)) to three named conditional lemmas (theorem-level subject to a separate H-spectral-gap assumption):

- **C2ca.1 Scalar Secular Cancellation:** algebraic secular identity proven; uniform decay $s_\lambda \rightarrow 0$ proven only under a separate Galerkin-truncation control. Numerical evidence (N -convergence-gated, 2026-06): $|\mu/\alpha| \approx 4\text{--}5 \times 10^{-7}$ across $\lambda = 3, 5, 7, 9$, stable in N (drifts $\leq 6\%$ up to $N=80$, digit-stable in precision) — i.e. a genuine finite Galerkin limit, *mildly increasing* in λ ($\approx 4.1 \rightarrow 4.9 \times 10^{-7}$). This supports *smallness*, not decay: the closure input $s_\lambda \rightarrow 0$ remains a purely analytical obligation.
- **C2ca.2 Projected Poisson Quasimode:** transfer formula $h = (1/n)P_0\Pi(E + B)$ proven; uniform analytical control of P_0 -leakage and boundary decay open. Numerical evidence: $p_\lambda \approx 2\text{--}3 \times 10^{-6}$.
- **C2ca.5 Coercive Complement:** finite Min–Max identity for a prescribed spectral window proven; the uniform order-one gap g_* is not derived from Cauchy-interlacing alone — it requires a separate H-spectral gap or fixed-waterline / slush-collar control (replaced by C2cf.1 *Fixed-Waterline Quasimode Projection*: in the finite self-adjoint Galerkin model, for any pre-chosen $g_0 > 0$, $\|(I - P_{\lambda, g_0})k\| \leq R_0/g_0$).

These three inputs plus the fixed-waterline assumption replace the earlier mass-adaptive gap rhetoric (which was diagnosed as circular against MS2, see Section 7).

3.3 Empirical Convergence and the Effective Microcluster

The mass-based effective cluster gap $g_{\text{eff}}(\varepsilon) := u_{J+1} - u_0$ with $J_\varepsilon := \min\{J : M(J) \geq 1 - \varepsilon\}$ stabilises at $g_{\text{eff}}(10^{-12}) \approx 5\text{--}6$ (order one, λ -independent), giving $R_0/g_{\text{eff}} \approx 5 \times 10^{-7}$. The earlier fixed-tolerance gap was unstable (cut inside the genuine cluster); the mass-based replacement is stable and macroscopic. See Zookeeper C2by/C2bz for the full analysis [9].

3.4 The Determinant Channel: Zero-Counting Reformulation and Normalisation Ledger

After the literal MS1 (uniformly simple cluster splitting) was excluded (Section 6), the working reformulation of MS1 is *zero counting*: compare the CCM secular function

$$F_N(z) = \sum_{j=-N}^N \frac{\xi_j}{z - p_j}, \quad p_j = \frac{2\pi j}{\log c},$$

(ξ = even ground state of the Galerkin matrix) against the completed Ξ via the argument principle on contours, instead of asking for eigenvalue simplicity. Writing $P(z) := F_N(z) \prod_j (z - p_j)$ for the degree- $2N$ secular polynomial, a dedicated measurement campaign (2026-06; scripts `ms1_det_zero_counting_harness.py`, `ms1_newton_power_sum_balance.py`, `ms1_cartwright_ledger.py`; full audit trail in the Zookeeper companion [9]) established the following RH-neutral structure ($c = 9, 14, 25$; $N \leq 48$; all statements numerical, no MS1/MS2 closure is claimed).

- **Root anatomy.** All $2N$ roots of P are real, in $\pm$ -pairs: K_h hit pairs matching γ_k to ≥ 30 decimal digits inside the comb range (earlier $\sim 10^{-15}$ reports were limited by hard-coded reference zeros, not by the roots), a graduated degradation governed by the comb reach $p_N = 2\pi N / \log c$ as the *only* empirically supported scale (a conjectured density-equality scale $T^* = 2\pi c$ left no signature in six dedicated runs), *no* comb-shadow roots, and a c -dependent but N -stable number of outer tail roots (5/7/10 pairs at $c = 9/14/25$).
- **Far field.** The tail roots carry the Newton power-sum defects exactly ($S_m(\text{tail})/c_m = 1.000$ for $m = 4, 6$ at all three c): the Hadamard envelope of Ξ is realised by P as a thinned outer node cloud (reminiscent of Gaussian-quadrature nodes; the analogy is used descriptively only).
- **Near field (exact ledger).** On a packet contour around $\{\gamma_1, \gamma_2\}$ the logarithm of the pairing ratio decomposes exactly into four blocks — hit mismatch, tail cloud, uncanceled zeros, comb — and the decomposition reconstructs every measured Rouché margin to 3–4 significant digits (35 configurations). The block hierarchy is $\text{comb} \gg \text{uncanceled zeros} \gg \text{tail} \gg \text{hit mismatch}$.
- **Canonical pairing.** Scanning all comb-compensation windows, the margin minimum sits at the *bare* pairing P vs. $C\Xi$ — no comb factor in the reference (margins 1.56/1.65/1.73 at $c = 9/14/25$, $N = 40$). Completing the comb to its closed form $\prod_{j \geq 1} (1 - z^2/p_j^2) = \text{sinc}(z \log c / 2)$ is the *worst* reference: its exponential type $(\log c)/2$ in $\text{Im } z$ clashes with the polynomial P — the Cartwright-type obstruction in pure form.
- **Local convergence rate.** The bare-pairing margin falls from 3.86 ($N = 16$) to 1.65 ($N = 40$) and is nearly c -independent at fixed N . Its source is almost exclusively the uncanceled zeros above the hit reach $T_{\text{hit}}(N)$ (≈ 50 at $N = 16$, ≈ 100 at $N = 40$); the measured spread ratio matches the prediction $\sigma_1(\gamma > T_{\text{hit}})$ from the Riemann–von Mangoldt density [11] within 1%.

Honest wall statement. The channel settles the normalisation question of the regularised-determinant formulation at finite N : the reference is the bare $C_N \Xi$,

and the local convergence $P_N \rightarrow C_N \Xi$ is quantified to first order with rate $\sigma_1(\gamma > T_{\text{hit}}(N)) \sim \log(T_{\text{hit}})/T_{\text{hit}}$. That $T_{\text{hit}}(N) \rightarrow \infty$ — the hyperconvergence of the secular roots onto the zeros — is exactly the unproven content of MS1; nothing in the ledger circumvents it. The open surface of the determinant channel is thereby concentrated in a single quantified rate.

4 Spectral Convergence Constants

Both spectral branches produce characteristic constants whose identification and saturation provide evidence for the conditional reductions. We collect them here for comparison; the cross-route synthesis with Branch A’s constant $C^* = 64\pi^2/3$ (Part II [7], § GF1) is deferred to Part IV [6].

4.1 Hadamard Curvature Constant c_Ξ (RH-neutral)

The Hadamard product for the completed Riemann ξ -function gives

$$\log \frac{|\Xi(it)|}{|\Xi(0)|} = \sum_{\rho} \log \left| 1 - \frac{-t^2}{(\rho - \frac{1}{2})^2} \right| \sim -\frac{t^2}{2} \sum_{\rho} \frac{1}{(\rho - \frac{1}{2})^2} + O(t^4),$$

whence

$$c_\Xi = -\frac{1}{2} \sum_{\rho} \frac{1}{(\rho - \frac{1}{2})^2} \approx 0.0231. \quad (6)$$

This is the leading quadratic curvature of $\log A_\Xi(\delta)$ near $\delta = 0$. The representation (6) sums over *all* non-trivial zeros and is *RH-neutral*. Under RH, $\rho = \frac{1}{2} + i\gamma_k$ and the sum specialises to $\sum_{k \geq 1} 1/\gamma_k^2$.

4.2 Residual Curvature κ_∞ (Branch B)

The multiplicative canonical form (3) decomposes A_λ into the Hadamard baseline A_Ξ and a residual factor $1 + \kappa(\lambda)\delta^2$. Empirically, $\kappa(\lambda)$ was initially reported as saturating at $[1.6, 1.8] \times 10^{-3}$; however, a dedicated N -convergence study (four-point exponential fit at $\lambda = 11$, $N \in \{137, 160, 200, 220\}$) shows $|\kappa_\infty(11)| \lesssim 10^{-4}$, approximately $15\times$ smaller. The earlier values conflated transient $\kappa(N)$ with the asymptotic limit. The sign of κ_∞ remains open ($\kappa_\infty(3) \approx -2.3 \times 10^{-3}$, negative and flat). As $\lambda \rightarrow \infty$, $\hat{\xi}_\lambda \rightarrow \Xi$ in the relevant topology; the residual curvature measures the finite- λ correction. The load-bearing requirement for Branch B (ii-a) is $\sup_\lambda |\kappa(\lambda)| < \infty$ (boundedness), not a specific sign.

4.3 Hecke-Normalised Gap $g_0(m)$ (Branch Z)

For the m -th Riemann zero γ_m , the classical Montgomery–Odlyzko pair-correlation heuristic [10] suggests a local mean zero spacing of $2\pi/\log \gamma_m$. The effective microcluster gap g_{eff} of Branch Z (Section 3.3) is consistent with this scaling: $g_0(m) \approx 2\pi/\log \gamma_m$.

Branch	Constant	Value	Status
B	c_{Ξ} (Hadamard curvature)	≈ 0.0231	analytical (RH-neutral)
B	κ_{∞} (residual curvature)	$ \kappa_{\infty}(11) \lesssim 10^{-4}$	corrected (preliminary $[1.6, 1.8] \times 10^{-3}$ was transient)
B	$B_{\Xi} = \xi(0)/\xi(1/2)$	$1.00579\dots$	analytical
Z	$g_0(m) \approx 2\pi/\log \gamma_m$	varies	classical (Montgomery–Odlyzko)

5 Common Wall and Spectral-Sector Outlook

The three branches share Wall (ii) as the common analytical obstacle, but attack it through structurally different mechanisms. The following comparison table summarises the conditional reductions.

Branch	Reduction chain	Open hypothesis (the wall here)	Status
A — Even Dominance	A1 + A2 + A7 $\Rightarrow$ RH	A7 (Asymp. Variational Gap Conj.) + A3 (Odd-sector lower bound)	conditional reduction
B — I-1 Normal-Family	B1 + B5 + B9 $\Rightarrow$ RH	(ii-a) (uniform strip bound from QW_N -coercivity)	conditional reduction
Z — CCM Microcluster	B1 + C2ca + fixed-waterline $\Rightarrow$ RH	C2ca.1 + C2ca.2 + C2ca.5 + $g_* \geq g_0$	conditional reduction

Shared inputs. A1 = B1 (both rest on Connes–vS Thm 5.6 [3]). The Foundation-Lock $\hat{\xi}(z) = (2/\sqrt{L})\sin(zL/2)F(z)$ from Branch B is the canonical criterion object used in Branch Z as well.

Distinct mechanisms. Branch A attacks Wall (ii) via eigenvalue ordering $\lambda_1^+ < \lambda_1^-$ (Part II [7]). Branch B attacks via a strip-uniform Hurwitz-style normal-family argument (this paper). Branch Z attacks via a microcluster quasimode-to-projector estimate (this paper + Zookeeper [9]).

Closest to conditional closure. Branch B currently has the smallest open surface (a single named hypothesis (ii-a)) together with strongly sub-generic empirical evidence ($A_{\lambda}(0.4) \approx 1.004$ flat over seven converged λ -points, Theorem B-1 giving $A_{\lambda} \leq 1.0063$ under the log-curvature hypothesis). Branches A and Z remain conditional reductions in the multi-gap sense (two or three open inputs respectively). The title of the present series follows the conservative convention: “Atlas” applies as long as no single branch has reached conditional-proof level under the audit protocol of the project `DECISION.md`.

Promotion protocol and future scenarios are discussed in Part V (Section 10 gives a brief forward reference).

6 Excluded Spectral Paths

The post-v2.0 spectral programme (Connes–van Suijlekom reformulation, I-1 normal-family reframe, err-collapse diagnostic) closed several routes. These are documented here for the audit trail; they do not affect the main conditional reduction.

- **Eigenvalue-ordering via a uniform spectral gap.** The plan to lift $\lambda_{\min}(W^+ - W^-) < 0$ to the eigenvalue inequality $\lambda_1^+ < \lambda_1^-$ via Kato–Temple, Davis–Kahan, or Lehmann gap estimates fails because the relevant minimum eigenvector of QW_N sits in a near-degenerate cluster (even/odd split $\sim 10^{-65}$ at $\lambda = 5$, $\sim 10^{-84}$ at $\lambda = 7$, super-exponentially collapsing with λ). No uniform spectral gap is available.

- **Localization hypothesis for the strip bound (ii-a).** A natural plan was to bound $|\hat{\xi}_\lambda(x + i\delta)|$ via localization assuming ξ_λ is concentrated near $t = 0$. This is falsified: ξ_λ is genuinely edge-concentrated ($r_{99} \approx L/2$, $|\xi_\lambda(0)| \approx 0$). A growth-adapted normalisation Φ is required.

- **Edge-profile factorisation for (ii-a).** The exact identity

$$F_\lambda(z) = 2 \cos(zL/2) C_\lambda(z) + 2 \sin(zL/2) S_\lambda(z)$$

isolates the explicit λ^δ -growth modulation. The edge route yields no type gain because the eigenfunction mass is spread ($r_{50} \approx 1$ uniformly, but $r_{99} \approx L/2$).

- **Literal MS1 (cluster ladder).** The conservative form of MS1 — uniformly simple splitting of the minimum eigenvalue of QW_N at every (λ, N) — fails: the splitting collapses super-exponentially with λ . Only a cluster/quotient reformulation has structural traction.
- **Strip-amplification at $\text{dps} = 30$ (retracted).** Early measurements at $\text{dps} = 30$ suggested sub-generic growth and non-monotone instability. At converged precision ($\text{dps} \approx 30\text{--}50 \cdot \lambda$), the picture is qualitatively different: $A_\lambda(0.4) \approx 1.004$ is flat. The methodological lesson (dps requirement scales linearly with λ for cluster-near-degenerate eigenvectors) is itself substantive.
- **Raw- L^2 Hurwitz passage (falsified 2026-05-15).** The plan to derive the Hurwitz strip-uniform convergence directly from a raw- L^2 tail estimate fails on two fronts: (a) the archimedean tail has norm $O(\lambda^{-1/2})$ — exactly boundary exponent $\alpha = 0$, no margin; (b) the prime-Dirac tail diverges in the $m = 1$ branch. Negative-Sobolev does not rescue this. The diagnostic identifies operator-norm topology (i.e. Branch B) as the only viable replacement. The Connes–Hurwitz–Passage project re-scoped on 2026-05-17 to a purely expository role (Geiger2026hurwitz, in preparation).

Remark 6.1 (Common pattern). These six exclusions share a structural pattern: the minimum-eigenvector cluster of QW_N is the source of the technical difficulty. Methods that assume or require a uniform gap, a thin-localised eigenfunction, or a generic function-class behaviour all fail. Productive routes work *with* the cluster (waterline / cluster-quotient reformulations, growth-adapted normalisations) rather than against it.

7 Excluded CCM Paths (Branch Z Heritage)

The CCM microcluster programme accumulated its own catalogue of dead routes. They do not affect the Branch-Z conditional reduction.

- **Mass-based microcluster definition.** Defining the microcluster via a mass criterion is circular against MS2. Replaced in Zookeeper v1.2+ by a *spectral* microcluster (low-energy eigenspace bounded by an outer gap g_* of order one).
- **Naive Hurwitz passage.** A direct Hurwitz application without approximation-quality control yields statements too weak for the err-collapse closure. Replaced by a Slepian–Pollak-/PSWF-based passage.

- **Anchor-perturbation / Banach contraction (C2br', falsified).** A Banach-norm contraction $C_{\max} < M$ fails by factor 55–7000 on all twelve tested configurations.
- **Projected frontier cancellation $S_{bulk} \approx -S_{bd}$ (C2bs, falsified).** Both contributions have the same sign and are both of size $\sim 10^{-7}$. The true cancellation mechanism sits in the eigenvalue equation $(A - w_0)k \approx 0$.
- **Uniform mass criterion C2ak.** The total off-cluster mass bound diverges with λ ($6.68 \rightarrow 15.3 \rightarrow 1865$ across $\lambda = 3, 5, 7$) because it discards internal cancellation.
- **Canonical local geometry C2al (directional defect).** The geometric inequality has the wrong direction — the correct form is $g_R/d_m \leq T_m^\beta/\beta_0$ (lower bound on d/g , not upper).
- **Cluster selection via Δ -projection.** Simple δ_N -projection is not a robust cluster selector: at $\lambda = 5, N = 55$ `best_zero=k=16` while the other two select $k = 17$.
- **Log-concavity / detailed balance (C2cc H1/H2/H4).** The hypotheses that β_n or $|b_n|$ are log-concave (H1), that the spectral residue chain has a detailed-balance structure (H2), or that a Mertens-anchor applies (H4) were tested at $\lambda = 3, N = 30, \text{dps} = 50$ and falsified. Structural cause: arithmetic (prime-driven) oscillation.
- **Direct Frontier-Dominance v2.1 (Killer 0, 2026-05-14).** The v2.1 argument used a common Rayleigh test vector to prove $\lambda_{\min}(D_{tot}) < 0$ and then identified $\Delta_{low} = \lambda_{\min}(D_{tot})$ with $\Delta = \lambda_1^+ - \lambda_1^-$. The reviewer counterexample $A = \text{diag}(0, 10), B = \text{diag}(5, -5)$ shows this does *not* follow. Even dominance for asymptotic λ remains conditional on a separate odd-sector lower bound. (This item is Even-sector-specific and documented in detail in Part II [7]; it is listed here because the falsification also restructured the comparative analysis of Branches B and Z.)
- **Euler-diagonal determinant (Z1, closed by Schatten trichotomy, 2026-06-09).** Realising the counting object as a regularised determinant over the Euler diagonal fails structurally: on $\text{Re } s = 1/2$ the diagonal lies in $\mathcal{S}_3 \setminus \mathcal{S}_2$ ($\det_2$ diverges via Mertens), while $\det_3$ loses the arithmetic $k = 1, 2$ kernel. Only a different singular-value class (Ruelle-transfer type) remains unexplored.
- **Contour counting inside Ω_E alone (F3, 2026-06-09).** Closing the zero count by contours lying entirely within the region of established convergence fails by a convexity lemma; any gain re-imports positivity/tightness a priori.
- **Constant- C packet Rouché / infinite-comb reference (2026-06-10/11).** On packet contours, Rouché domination against a constant multiple of Ξ fails in the $N \rightarrow \infty$ limit at all tested c (margin plateau 1.541 at $c = 9$; rising through ≈ 0.95 at $c = 14$); completing the comb to its sinc closed form makes the reference strictly worse (exponential-type clash). The z -dependent normalisation question is settled in Section 3.4.
- **Root-pairing bulk models (B_K , PolyB, 2026-06-09/10).** Modelling the bulk factor by pairing outer secular roots against outer zeros is inert (the roots hit the zeros below measurement floor, $B_K \equiv 1$); the bare local root polynomial against

$C\Xi$ fails by two orders of magnitude (margin 35). The bulk is carried exclusively by the compensation polynomial and the Ξ envelope.

Remark 7.1 (CCM-pattern: cancellation, not magnitude). The CCM-side dead routes share a different pattern from the spectral side: each tries to bound a quantity by its individual-term magnitude, and each is defeated by large internal cancellation. The living routes explicitly factor the cancellation through projector identities or scalar secular equations, never through simple norm domination.

8 Methodical Dead Ends

A small number of methodical routes were tested on the spectral side and ruled out on structural grounds.

- **Dobrushin–Zegarlinski uniqueness $\rightarrow$ RH-CAP.** Dobrushin–Zegarlinski uniqueness works only on lattices with localised Gibbs measures. RH has neither; the absence of multiplicative independence enforces a non-local prime-coupling structure that breaks the Dobrushin contraction setting.
- **Sturm node-counting (naive).** Classical Sturm–Liouville node-counting for the eigenfunctions of QW_N fails because the operator is finite-dimensional and the even/odd spectral gap is sub-exponentially small. Node-counting provides no information at the relevant precision scale.
- **Naive Abel summation for far-tail.** Applying Abel summation to the far-tail prime sums $\sum_{p>\lambda} (\log p)^k / p^s$ without controlling the arithmetic oscillation of $\theta(x) - x$ leads to divergent error terms. The PNT transfer lemma (Part II) shows how to do this correctly.
- **Naive superdeterminant form for Möbius- $\mathbb{Z}_2$.** The formal identity $\text{sdet}(I - L_s) = \det(I - L_s^+) / \det(I - L_s^-) = 1/\zeta(s)$ is algebraically correct for finite prime sets but does not converge for the full prime set at $\Re(s) = \frac{1}{2}$ (trace-class obstruction K4 from Part I).
- **Friedrichs + Petersson as Hilbert–Pólya lever.** The plan to use Friedrichs extension of the Petersson inner product to construct a self-adjoint realisation whose spectrum encodes the Riemann zeros fails: the relevant bilinear form is not semi-bounded on the domain needed to capture the critical-line zeros.

9 Update (v3.2): Closed Units, June–August 2026

Between June and August 2026 a series of finite, audit-carrying working units was closed on and around the spectral sector. They are recorded here in the same spirit as the exclusion catalogues above: each entry states what was examined, how far it was verified, what it means for the route, and — explicitly — what it does *not* claim. None of them closes RH, and none changes the conditional-reduction status of Branches B and Z established in the preceding sections.

Remark 9.1 (Reading the counts in this section). Two kinds of ratio appear below and must not be read alike. A *verification quota* — “producer 18/18”, “independent verifier 22/22” — reports checks that were carried out and passed. A *shortfall count* reports how many required items are *absent*: where the text below says that none of fifteen prerequisites is met, that none of fourteen proof-carrying categories is present, or that none of twenty-one packages exists, these are missing-item statements and emphatically not pass rates. For that reason every shortfall is written out in words, while genuine verification quotas are given as ratios and named as such.

9.1 Block A: The Freedman/Weyl Continuum Route

- **Quotient-to-Weyl commitment contract V1 (exact finite contract with counter-model).** The unit fixes contractually what a lift from finite quotient positivity to the Weyl/continuum statement must deliver, and secures this by a counter-model that blocks the inference from positivity on $\ker R$ to full positivity. Verification: producer 14/14, independent 16/16, with a byte-level attack on the bound sources failing closed. A later reading of the project register (state V1.1, more recent than the plan state) sharpens the picture: the source discharges the boundary obligations itself — $\beta_{\text{bdy}} = 0$ and $D_{\text{bdy}} = 0$ by a primitive endpoint computation, and $D_q|_Y = 0 \iff D_q = 0$ on X_R by density — and states the equivalence $D_q = 0 \iff \Gamma^*\Gamma \leq C \iff Q_\Phi \geq 0$ as a direct lift theorem of the original. For the route map this means that, on register state V1.1, the Weyl lift is no longer a diffuse list of seven continuum obligations, but is concentrated on a single analytical residual obligation: $\Gamma^*\Gamma \leq C$ on X_R , uniformly in ω , together with the framing obligations. The finite quotient finding alone still does not carry the lift. Claim boundary: no PASS is asserted for $\Gamma^*\Gamma \leq C$, for the lift, for Weyl, for KLM or for RH, and the earlier formulation “seven continuum obligations open” is superseded by register state V1.1.
- **Uniform- ω reweighting audit V1 (finite negative finding).** Positive and locally uniformly integrable branch weights do *not* preserve the aggregated square domination without an additional hypothesis. Verification: producer 15/15, independent 17/17, fail-closed. Because the finding is finite and exact rather than numerical evidence, it is usable directly as a route decision. Its consequence is that the obvious reweighting strategy for $|\omega| < 1/2$ dies; what remains open is the operator-bound alternative (branchwise domination, a uniformly weighted intertwiner, a conjugated contraction, and a parameter cover with positive margin). Claim boundary: this is *no* counterexample to the actual Freedman kernel — what is refuted is the strategy, not the statement.
- **Endpoint-to-continuum follow-on chain validation and DAG audit (negative).** Eleven dashboard rows were reduced to four source-coherence cuts, and the follow-on chain endpoint $\rightarrow$ rank $\rightarrow$ Green $\rightarrow$ unique continuation $\rightarrow$ active range was validated negatively. The bound sources carry no mutation counts for this unit, and none are supplied here. The result is not a universal no-go: it is a negative finding for exactly this chain and for no wider claim. As a by-product, the orphan-file cross-reference was closed as a finite diagnostic instrument. Claim boundary: neither the endpoint layer nor the continuum target is refuted; what fails is the specific chaining of the two.

- **All-time/continuum and pullback readiness audit V1 (fail-closed).** Fifteen prerequisites of an all-time-capable, continuum-capable and non-circular KLM $\rightarrow$ de Branges route were audited, and *none of the fifteen is met* — a shortfall count in the sense of Theorem 9.1, not a pass rate. The audit itself is clean and carries its own quotas: producer 18/18, independent verifier 22/22, terminal status NOT_READY_MISSING_BOUND_CONTINUUM_AND_PULLBACK_OBJECTS_NEW_MATHEMATICAL_INPUT_REQUIRED. In this path no autonomously executable static or finite computational step remains, so movement requires new mathematical input rather than further computation. Claim boundary: the bridge is unfinished, not refuted — the audit measures readiness, not truth.

9.2 Block B: The KLM/de Branges Route

- **KLM $\rightarrow$ de Branges pullback-obligation audit V1 (cross-term counter-model).** From separately nonnegative tail and boundary differences, full signed Gram positivity does *not* follow. Verification: producer 17/17, independent 19/19, fail-closed. The route consequence is that the “separately nonnegative suffices” strategy dies, and the untreated cross terms are thereby identified as the decision locus for any successor attempt. Claim boundary: the actual shifted- Ξ kernel is not refuted; what falls is the sufficiency argument, and it falls to an explicit counter-model rather than to a numerical margin.
- **KLM support, isometry and defect-lift layer (four closed findings).** On the positive side, a weighted full-line support H_ρ realises both Mellin halves (producer 28/28, independent 34/34), and a target-independent fixed-trace support Mu was constructed (producer 19/19; verifier with 49 direct checks plus 21/21). On the negative side, no scalar $a_t T_t$ is isometric for $t \neq 0$ and an exact weight cocycle destroys the k_z eigenrelation (producer 22/22, independent 28/28); a universal joint cross-Gram repair of the defect lifts $J_{(\sigma,u)}$ is refuted (producer 26/26, independent 31/31, with $126 + 378$ controls); and the (Λ_a, Mu_z) nullspace annihilation is vacuous from saturation onwards (producer 17/17, independent 21/21). The route reading is that the supports exist, while the isometry route and the universal defect-lift repair route are closed — which also devalues an earlier item that had been recorded as progress. Claim boundary: each negative statement is bound to its stated class (scalar isometries, exact cocycles, universal joint repairs), and none of them refutes the KLM pullback target itself.
- **Complete normalised KLM residual block — directed non-PSD (final finding).** Five preregistered Hardy nodes and all ten Riesz vectors were taken from directed normalised 79×10 branch maps, and the Hardy target block was re-enclosed with Arb/Acb (20 modes, majorants below 2^{-120}). For the residual block $\Delta = H - C$ all ten real diagonal boxes have strictly negative upper bounds, the largest being $\approx -0.2665171987\dots$. Verification: 26/26 sources byte-bound, 384-bit producer 17/17, independent 512-bit verifier 20/20, fail-closed. This joint branch-symmetric five-node polar and graph normalisation therefore dies rigorously as a positive pullback candidate, as an endpoint finding rather than a provisional one; two coarser variants had already been excluded numerically beforehand (plus-matching $(5, 0, 5)$ with $\lambda_{\min} \approx -0.11543$, unit direct sums $(4, 1, 5)$ with $\lambda_{\min} =$

−12.2767...). Claim boundary: the result binds this one finite candidate; the Mellin split is not refuted by it.

- **Operator-semantic alternatives audit V1.** Three admitted classes were examined for a target-free alternative: branch-weighted graph metrics, fixed trace augmentations, and coupled defect channels. None of the three yields a unique target-free choice, and no matrix was produced. Verification: producer 15/15, verifier 17/17, terminal status `NO_TARGET_FREE_UNIQUE_FINITE_ALTERNATIVE_MATERIALIZED_SWITCH_TO_CONTINUUM_PULLBACK`. The route consequence is that the finite matrix path is exhausted, which is what routes this line into the readiness audit of Block A above. Claim boundary: exhaustion is asserted for the three admitted classes only — a class outside that admission is untested rather than excluded.

9.3 Block C: Weil/CvS Numerics

- **RH-V4R foreign-operator run (terminal, independently verified).** One authorised run of a foreign operator returned code 0 with checkpoints at $N = 48/56/64$. Its terminal result is `C13_FIXED_LADDER_COMPLETE_TAIL_GATES_NOT_REACHED`: the Rayleigh tail passes, the fit tail fails ($0.1273... > 0.0005$). Verification was independent on two levels — the result verifier discarded 10/10 mutations, the terminal-envelope verifier 17/17, including a checkpoint byte mutation. The route reading is that this line has been measured, not confirmed; the open item `CVS_N_RESOLUTION_UNRESOLVED` stands unchanged. Claim boundary: the failed fit tail is a gate result rather than a refutation of the underlying route, and no retry is admissible without a new contract.
- **K1 scalar audit, transitive source audit, and fail-closed full-replay input contract.** K1 remains open at the level of causes. The scalar audit (256-bit Arb) gives a fit-error quotient of $3.9533...$, while the Rayleigh side spans only 1.4489×10^{-8} . The transitive source audit shows that the wrapper does execute the V4 kernel but leaves a provenance gap, since path and SHA are missing from the manifests — a historical defect that does not invalidate the run automatically. A fail-closed full-replay input contract now specifies exactly fourteen proof-carrying categories for $N = 40, \dots, 64$, and *none of these fourteen categories is currently present*; in the sense of Theorem 9.1 the fourteen are the missing items, not passed checks. Claim boundary: the unit prices the resumption of K1 exactly, but it neither resumes it nor upgrades any claim.
- **Matrix production run D-20260813-003 (terminally verified, candidate missing).** A singly authorised Arb matrix production run ($c = 13$, $N = 48/56/64$, $T = 200$, 3225.75 s) returned code 0 and passed verification. The verifier reconstructed the 49×49 , 57×57 and 65×65 matrices, confirmed 10/10 prime components and discarded 12/12 mutations; the package is byte-identical (SHA 78ba72b5...aab0e7). The terminal status is `RIGOROUS_MATRIX_PRIMITIVES_READY_CANDIDATE_MISSING`, recorded with `candidate_verified_here=false` and `claim_upgrade=false`. What the run delivers is rigorous matrix primitives; what it does not deliver is the candidate — candidate realisation, angle control, and the sector condition $\mu_{1,g^+} < \min(\mu_{2,g^-}, \mu_{1,u^-})$ are separate gates that were not executed

here. Claim boundary: primitives ready is not candidate verified, and the run explicitly upgrades no claim.

9.4 Block D: Diagnostics and Hygiene

- **CCM–Groskin metadata package.** Twenty-one normalised cells of the metadata package were audited. The bound sources record no producer or verifier quotas for this unit, and none are supplied here. The package is explicitly `DIAGNOSTIC_ONLY_NOT_VALIDATED`: it carries diagnostic value but no evidence level, and its negative blocks are warning signals rather than demonstrated spectral pollution. Open items include the $c = 100$ runner binding, the raw spectra, the smallest-positive branch selection, the classification, the spectral-pollution test, the operator realisation, the limit topology, and the $c/N/T$ transition. Claim boundary: nothing in this package supports or refutes a spectral statement, and reading its negative blocks as established pollution would over-read the material.
- **Residue/determinant hygiene gate (three-axis label audit).** A three-axis label audit covering 13/13 claim lines, 7/7 active files and 21/21 axes, with 14/14 independent mutations discarded, established a bookkeeping rule that has kill effect. The rule keeps three channels strictly separate: the numerical eigenresidue, the meromorphic zero residue, and the determinant channel of Section 3.4. New spectral files must continue this contract rather than restate it. Its operative consequence is the distinction that a technical PASS is not an analytical channel PASS — a label conflating the three is void under the gate regardless of the numbers behind it. Claim boundary: the gate is a hygiene instrument, so it decides the admissibility of labels and not the truth of any spectral statement.

9.5 Active Fronts (Status Lines Only)

The following fronts are open at the time of this update. Each is given one status line and no further detail; the working registers carry the substance.

- **Congruence reduction / residual gate G-A/G-B.** The structure theorem is available and the gate is defined but not executed, and a new contract with directed arithmetic is required.
- **21-node production run.** None of the twenty-one packages exists, and a separate production contract and authorisation are outstanding (the design uses 21 collocation nodes at order 200, not 200 nodes).
- **$N1/\Phi_T$ front.** A separate front with its own register; not summarized here.
- **KLM branch maps.** Advanced as far as the normalised residual finding above, with no autonomously executable step available without new mathematical construction input.

10 Summary

Contribution	Status	Location
I-1 Normal-Family Reframe: (ii) $\rightarrow$ (ii-a) + (ii-c)	rigorous	Section 2.2
(ii-c) absorbed into (ii-a) under (Z)	rigorous	Section 2.2
Foundation-Lock: canonical criterion object	rigorous	Section 2.2
Converged empirics: $A_\lambda(0.4) \approx 1.004$ (7 points)	numerical evidence	Table 1
Multiplicative canonical form	empirical (0.012% fit)	Section 2.4
Theorem B-1: $A_\lambda \leq 1.0063$ (conditional)	conditional	Theorem 2.3
c_Ξ Hadamard curvature (RH-neutral)	analytical	Section 4
CCM: three C2ca-inputs identified	conditional reduction	Section 3.2
CCM: effective microcluster gap $g_{\text{eff}} \approx 5\text{--}6$	numerical evidence	Section 3.3
Determinant channel: exact ledger, bare $P \leftrightarrow \Xi$ pairing, rate $\sigma_1(\gamma > T_{\text{hit}})$	numerical evidence	Section 3.4
6 excluded spectral paths + 13 excluded CCM paths	documented	Sections 6 and 7
5 methodical dead ends	documented	Section 8

The spectral sector contributes two independent conditional reductions of RH, each with a well-characterised open surface. Branch B has the smallest open surface (a single hypothesis (ii-a)) and the strongest empirical support ($A_\lambda \leq 1.0063$ conditional on uniform log-curvature control). The cross-route synthesis across all three branches is developed in Part IV [6]; the condensed atlas and outlook in Part V [5].

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